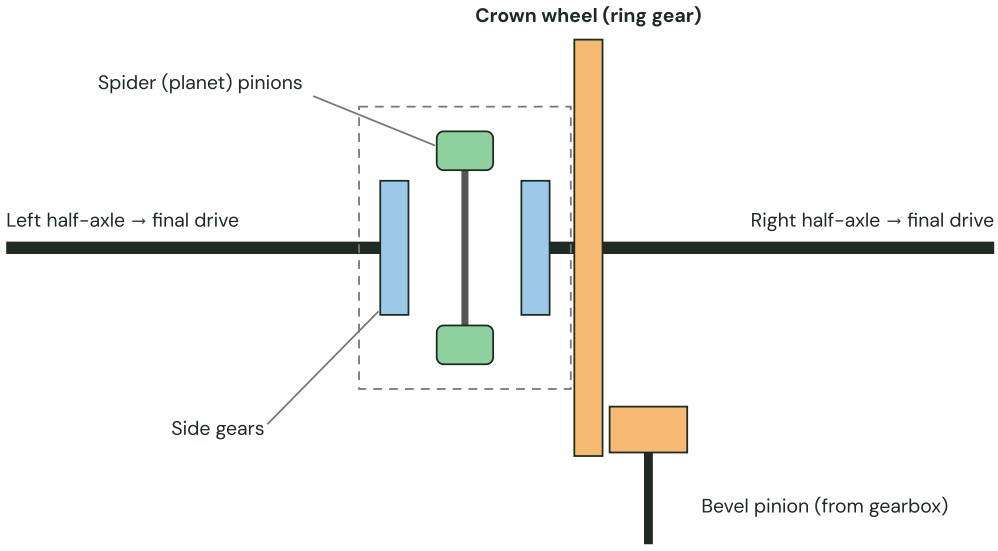
UNIT 4 · TOPIC 5(C)

Differential, Final Drive and Steering Brakes

Syllabus topic: Tractor power transmission: clutch, gearbox, differential and final drives



Differential unit — also gives right-angle drive and reduction (redrawn from the source figure)

How the differential works

- The bevel pinion + crown wheel give a right-angle drive AND a large reduction.
- The spider (planet) pinions in the cage let the half-axles turn at different speeds on turns while sharing torque EQUALLY.

$$N_L + N_R = 2 \cdot N_{crown}$$

Example. Crown wheel 100 rpm, inner wheel 85 rpm → outer = 2 × 100 - 85 = **115 rpm**.

Weakness — and the differential lock

- Because torque splits equally, if one wheel loses grip (mud) the OTHER also loses drive — torque is limited by the slipping wheel, which spins at double speed.
- **Remedy:** the DIFFERENTIAL LOCK (foot-operated dog clutch locking both axles) or a limited-slip differential.
- **Disengage the diff lock before turning!**

Final drive

- The last reduction (spur pinion–bull gear, or a planetary hub) sits at or near the wheels, so the differential and axle shafts carry LOW torque.
- It also raises ground (crop) clearance.

Independent steering brakes

- They act on the bull-pinion/differential shafts of each rear wheel.
- Pressing the inside-wheel brake tightens a turn.
- Latch both pedals together for road travel.

Practice MCQs

Q121. The crown wheel and bevel pinion also provide:

(a) Straight drive
(b) Right-angle drive with large reduction
(c) PTO drive
(d) Steering

Q122. The differential speed relation is:

(a) $N_L - N_R = N_{crown}$
(b) $N_L + N_R = 2N_{crown}$
(c) $N_L \times N_R = N^2$
(d) $N_L = N_R$ always

Q123. Crown wheel 100 rpm and inner wheel 85 rpm → outer wheel:

(a) 100
(b) 105
(c) 115
(d) 130 rpm

Q124. A standard differential splits TORQUE between half axles:

(a) By speed
(b) Equally
(c) To the faster wheel
(d) To the heavier wheel

Q125. When one drive wheel spins in mud, the differential:

(a) Sends full torque to the gripping wheel
(b) Limits both wheels to the slipping wheel's torque
(c) Locks itself
(d) Stops the engine

Q126. The differential lock:

(a) Stops the tractor
(b) Couples both half-axles to rotate together
(c) Frees the PTO
(d) Applies brakes

Q127. Before turning at headland, the diff lock must be:

(a) Engaged
(b) Disengaged
(c) Oiled
(d) Doubled

Q128. The final drive's purposes include:

(a) Final large reduction near wheels and reduced load on diff shafts
(b) Cooling
(c) Fuel metering
(d) Suspension

Q129. Planetary final drives are located:

(a) In the gearbox
(b) In the wheel hubs
(c) On the PTO
(d) At the radiator

Q130. Tractor steering brakes act on the:

(a) Front wheels
(b) Bull-pinion/differential shafts of each rear wheel
(c) PTO
(d) Crankshaft

Q131. For road transport, brake pedals must be:

(a) Independent
(b) Latched (locked) together
(c) Removed
(d) Disconnected

Answers

Q121 — (b) Right-angle drive with large reduction

Q122 — (b) $N_L + N_R = 2N_{crown}$

Q123 — (c) 115 — *$N_{outer} = 2 \times 100 - 85 = 115$ rpm.*

Q124 — (b) Equally

Q125 — (b) Limits both wheels to the slipping wheel's torque — *The spider gears balance torque — grip is limited by the WORSE wheel; hence the diff-lock pedal in wet fields.*

Q126 — (b) Couples both half-axles to rotate together

Q127 — (b) Disengaged

Q128 — (a) Final large reduction near wheels and reduced load on diff shafts

Q129 — (b) In the wheel hubs

Q130 — (b) Bull-pinion/differential shafts of each rear wheel

Q131 — (b) Latched (locked) together

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; IS 4931/ISO 500 (PTO), IS 4468/ISO 730 (linkage); NPTEL farm power lectures.